

## VESYOLAYA GORKA, Russian Federation

WMO: 314180

Lat: 52.283N

Lon: 135.800E

Elev: 659

StdP: 14.35

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -26.7      | -23.6 | -33.6                         | 0.8 | -24.0 | -30.1 | 1.0 | -21.8 | 14.3                  | 9.9  | 12.2 | 13.1 | 0.5                   | 20   | 0.571 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.0                   | 84.7            | 68.5 | 81.0 | 66.8 | 77.6 | 65.1 | 71.7                | 80.9 | 69.4 | 77.7 | 67.3 | 74.7 | 3.3                  | 140  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.6                            | 107.8 | 76.7 | 66.4 | 99.5 | 73.8 | 64.4 | 92.8 | 71.7 | 36.0          | 81.2 | 34.0 | 78.0 | 32.2 | 74.6 | 92.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |
| 11.9              | 9.1  | 7.2 | -29.6                      | 90.8 | 4.2                | 3.9 | -32.6                                              | 93.7 | -35.1      | 95.9 | -37.4      | 98.1 | -40.5      | 101.0 |
|                   |      |     | -29.9                      | 76.0 | 4.1                | 4.3 | -32.8                                              | 79.1 | -35.2      | 81.6 | -37.5      | 84.1 | -40.5      | 87.2  |

## Monthly Climatic Design Conditions

|       |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|-------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|       |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)   | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 31.9  | -7.0  | 0.8   | 17.1  | 35.0  | 48.6  | 59.7  | 65.6  | 63.6  | 53.1  | 36.7  | 13.0  | -5.4  | (6)  |
| (7)   |                                                                   | DBStd       | 27.45 | 10.35 | 8.94  | 9.10  | 8.15  | 8.72  | 8.05  | 6.77  | 5.83  | 6.90  | 7.70  | 11.90 | 10.64 | (7)  |
| (8)   |                                                                   | HDD50       | 8059  | 1766  | 1378  | 1020  | 457   | 136   | 10    | 0     | 1     | 46    | 418   | 1110  | 1717  | (8)  |
| (9)   |                                                                   | HDD65       | 12276 | 2231  | 1798  | 1485  | 899   | 513   | 199   | 75    | 97    | 359   | 878   | 1560  | 2182  | (9)  |
| (10)  |                                                                   | CDD50       | 1448  | 0     | 0     | 0     | 7     | 91    | 301   | 483   | 421   | 140   | 5     | 0     | 0     | (10) |
| (11)  |                                                                   | CDD65       | 192   | 0     | 0     | 0     | 0     | 4     | 40    | 93    | 52    | 3     | 0     | 0     | 0     | (11) |
| (12)  |                                                                   | CDH74       | 1662  | 0     | 0     | 0     | 1     | 79    | 456   | 752   | 345   | 28    | 1     | 0     | 0     | (12) |
| (13)  |                                                                   | CDH80       | 429   | 0     | 0     | 0     | 0     | 18    | 122   | 221   | 66    | 2     | 0     | 0     | 0     | (13) |
| (14)  | Wind                                                              | WSAvg       | 2.4   | 1.8   | 2.3   | 3.2   | 3.3   | 3.1   | 2.3   | 1.9   | 2.0   | 2.3   | 2.6   | 2.4   | 1.9   | (14) |
| (15)  | Precipitation                                                     | PrecAvg     | 26.0  | 0.7   | 0.5   | 0.9   | 1.6   | 3.1   | 3.1   | 4.3   | 4.1   | 3.3   | 2.4   | 1.1   | 1.0   | (15) |
| (16)  |                                                                   | PrecMax     | 33.3  | 1.7   | 2.0   | 2.6   | 4.5   | 5.7   | 6.2   | 8.0   | 8.7   | 7.1   | 7.7   | 3.0   | 3.1   | (16) |
| (17)  |                                                                   | PrecMin     | 14.5  | 0.0   | 0.0   | 0.0   | 0.2   | 0.6   | 0.8   | 0.1   | 1.5   | 0.4   | 0.6   | 0.0   | 0.0   | (17) |
| (18)  |                                                                   | PrecStd     | 5.1   | 0.5   | 0.4   | 0.7   | 1.0   | 1.4   | 1.3   | 2.2   | 2.0   | 1.8   | 1.7   | 0.9   | 0.9   | (18) |
| (19)  | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 23.5  | 28.3  | 45.2  | 68.3  | 82.5  | 88.5  | 91.7  | 86.2  | 77.4  | 64.4  | 42.7  | 27.8  | (19) |
| MCWB  |                                                                   |             | 21.3  | 23.4  | 34.6  | 49.8  | 59.4  | 68.8  | 73.6  | 70.4  | 62.4  | 49.7  | 34.1  | 26.1  | (20)  |      |
| 2%    |                                                                   | DB          | 19.3  | 22.4  | 39.1  | 60.1  | 74.7  | 83.5  | 85.3  | 81.3  | 72.8  | 56.5  | 36.0  | 22.1  | (21)  |      |
|       |                                                                   | MCWB        | 17.3  | 18.7  | 31.0  | 45.5  | 55.6  | 65.4  | 70.5  | 68.2  | 60.0  | 45.2  | 29.8  | 19.8  | (22)  |      |
| 5%    |                                                                   | DB          | 14.1  | 17.8  | 34.5  | 54.1  | 70.0  | 79.2  | 81.7  | 77.8  | 68.7  | 51.8  | 32.2  | 16.0  | (23)  |      |
|       |                                                                   | MCWB        | 12.4  | 14.8  | 27.8  | 41.6  | 53.3  | 63.4  | 68.7  | 66.5  | 57.8  | 41.9  | 28.2  | 14.1  | (24)  |      |
| 10%   |                                                                   | DB          | 9.2   | 14.1  | 30.7  | 48.7  | 65.2  | 75.1  | 77.9  | 74.4  | 64.6  | 47.9  | 29.4  | 10.1  | (25)  |      |
|       |                                                                   | MCWB        | 7.5   | 11.6  | 25.1  | 38.0  | 50.9  | 61.8  | 67.4  | 64.8  | 55.5  | 39.9  | 25.7  | 8.3   | (26)  |      |
| (27)  | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 22.0  | 23.3  | 35.1  | 50.7  | 60.8  | 72.2  | 76.3  | 73.5  | 65.0  | 50.6  | 34.7  | 25.9  | (27) |
| MDCB  |                                                                   |             | 23.7  | 28.2  | 44.1  | 65.4  | 76.1  | 84.2  | 86.4  | 81.6  | 73.5  | 62.9  | 41.0  | 27.4  | (28)  |      |
| 2%    |                                                                   | WB          | 17.3  | 18.7  | 31.3  | 45.9  | 57.2  | 68.0  | 72.7  | 70.6  | 62.2  | 46.5  | 31.0  | 19.7  | (29)  |      |
|       |                                                                   | MDCB        | 19.2  | 22.3  | 38.3  | 60.0  | 72.3  | 79.0  | 82.4  | 78.1  | 70.0  | 54.4  | 34.6  | 21.6  | (30)  |      |
| 5%    |                                                                   | WB          | 12.4  | 15.0  | 28.2  | 41.7  | 54.2  | 65.4  | 70.4  | 68.4  | 59.4  | 43.7  | 28.4  | 14.0  | (31)  |      |
|       |                                                                   | MDCB        | 14.0  | 17.9  | 34.0  | 53.5  | 68.0  | 75.8  | 79.3  | 75.1  | 65.7  | 50.5  | 32.1  | 16.0  | (32)  |      |
| 10%   |                                                                   | WB          | 7.5   | 11.6  | 25.6  | 38.2  | 51.5  | 62.9  | 68.3  | 66.5  | 57.1  | 40.8  | 25.8  | 8.4   | (33)  |      |
|       |                                                                   | MDCB        | 9.1   | 14.0  | 30.4  | 47.9  | 63.2  | 72.6  | 76.1  | 72.4  | 62.6  | 46.8  | 29.2  | 10.1  | (34)  |      |
| (35)  | Mean Daily Temperature Range                                      | MDBR        | 12.4  | 14.9  | 16.6  | 16.5  | 19.6  | 20.1  | 17.0  | 16.7  | 18.3  | 14.6  | 12.7  | 12.1  | (35)  |      |
| 5% DB |                                                                   | MCDBR       | 13.6  | 16.4  | 19.5  | 25.0  | 28.9  | 27.2  | 24.3  | 23.0  | 24.1  | 21.0  | 13.6  | 13.2  | (36)  |      |
|       |                                                                   | MCWBR       | 12.2  | 13.4  | 13.4  | 15.0  | 14.6  | 13.1  | 11.9  | 11.6  | 12.7  | 12.7  | 10.0  | 12.1  | (37)  |      |
|       |                                                                   | MCDBR       | 13.0  | 16.2  | 18.5  | 24.5  | 27.2  | 24.2  | 21.5  | 19.7  | 21.2  | 19.3  | 12.5  | 13.2  | (38)  |      |
| (39)  |                                                                   | 5% WB       | MCWBR | 11.8  | 13.3  | 12.8  | 14.9  | 14.3  | 13.0  | 11.9  | 11.2  | 12.2  | 12.2  | 9.4   | 12.1  | (39) |
| (40)  | Clear-Sky Solar Irradiance                                        | taub        | 0.256 | 0.287 | 0.353 | 0.460 | 0.536 | 0.487 | 0.490 | 0.415 | 0.365 | 0.363 | 0.297 | 0.254 | (40)  |      |
| (41)  |                                                                   | taud        | 2.139 | 2.088 | 1.940 | 1.818 | 1.812 | 2.059 | 2.098 | 2.325 | 2.398 | 2.187 | 2.112 | 2.101 | (41)  |      |
| (42)  |                                                                   | Ebn at Noon | 233   | 258   | 258   | 242   | 230   | 246   | 242   | 257   | 258   | 225   | 208   | 207   | (42)  |      |
| (43)  |                                                                   | Edh at Noon | 22    | 32    | 46    | 60    | 63    | 50    | 47    | 36    | 30    | 30    | 23    | 19    | (43)  |      |
| (44)  | All-Sky Solar Radiation                                           | RadAvg      | 449   | 821   | 1329  | 1658  | 1570  | 1698  | 1542  | 1335  | 1040  | 652   | 438   | 340   | (44)  |      |
| (45)  |                                                                   | RadStd      | 15    | 26    | 57    | 108   | 172   | 179   | 154   | 137   | 98    | 50    | 32    | 17    | (45)  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)  | (e)                    | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) | Station Only           | +0.70 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (9 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page